The Substitution State Machine: Monotonicity, Monotone Reads, Clone Stability, and Rollback

Deep dive E for "Refinement and Inference Growth in SKJS"

Models revision `bafdcf7c` of `main` (2026-08-05)

Abstract

The main paper models the inference substitution as a state machine; four of its ledger rows concern that machine's state discipline — axiom **L-MONO** (refinement-time monotonicity), the obligation *Monotone Reads Under Growth*, assumption **L-CLONE** (clone stability), and the obligation *No Commitment on Failure*. We complete the state model with the version counter, four caches, cycle-detection sets, recursive-subtyping depth counter, constraint attachment, skolem-identifier supply, and candidate-freezing flag, then analyze the four rows together.

The implementation fact that makes monotonicity possible is that `TVar::setConstraint` returns a new frozen value rather than mutating shared state. Under the two conditions stated in the paper, legal state transitions therefore preserve membership in types computed earlier (**L-MONO**). The audit of later reads finds two exceptions: a `seenSub` result recorded while candidate growth is suppressed can be reused after the candidate grows (**MR-1**), and the reflexive `TSInst` fast path reads an unsealed candidate through `peekRaw` (analyzed in deep dive D).

For cloned substitutions, the only refinement probe that generates inference variables consumes them entirely within temporary clones; no generated variable reaches a type installed against the main substitution. This proves **L-CLONE** subject to the fork-state form of **L-INTER**. Finally, an exhaustive save/restore review establishes No Commitment on Failure, with one separately justified exception: array-literal commitments intentionally survive some failed checks.

Contents

1 Purpose and Rows Addressed **2**

2 The Complete State Model **2**

2.1 Saved state: `SubstState` . . . 2

2.2 Unsaved state . . . 3

2.3 Constraint attachment is a value operation . . . 4

2.4 The extended transition table . . . 5

3 The Order-Theoretic Core **6**

4 L-MONO: Refinement-Time State Monotonicity **7**

4.1 Enumeration of refinement-triggered evolutions . . . 7

4.2 Discharge . . . 8

5 Monotone Reads Under Growth **8**
5.1 The invariant 8
5.2 Site audit 9
5.3 Residual hazard: the freeze-window unsealed read 9

6 L-CLONE: Clone–Compute–Install **10**
6.1 The pattern and what is actually lost 10
6.2 Over-approximation: the positive result 11
6.3 Clone-escaped fresh identifiers: an unreachable hazard 11

7 No Commitment on Failure: the Rollback Audit **12**
7.1 Exhaustive save/restore pairing 12
7.2 The intentional exception: array-literal commitments 13
7.3 `tenv.seenSub`: the #446 argument, made precise 14

8 Status Table and Ledger Edits **14**

A Implementation Anchors **17**

1 Purpose and Rows Addressed

This document supports the "Assumptions and Proof Obligations" ledger of the main paper. It addresses four rows — **L-MONO**, Monotone Reads Under Growth, **L-CLONE**, and No Commitment on Failure — treated in Sections 4, 5, 6, and 7, and summarized row by row in §8.

All file and line references are against commit `bafdcf7c`. Line numbers are cited as `file:line`; they identify the revision precisely and will drift with edits.

2 The Complete State Model

The main paper's Definition "Substitution States" records only the binding map and the freeze flag. The implementation state is larger. This section is the reference model; everything later in the paper is proved against it.

2.1 Saved state: `SubstState`

Definition 1 (Full Substitution State)**.** *A substitution state ρ consists of the fields of the Skip class `SubstState` (`TTypeSubst.sk:13-49`):*

- values : TVarID ⇀ SVOpen(TType) | SVClosed(TType) — *the binding map of the main paper's transition table;*
- version ∈ ℕ — *a counter with two duties: (a) it is incremented immediately before* every *write to* values *(`grow`, `TTypeSubst.sk:200`; `get`'s two sealing writes, `:214` and `:224`); and (b) it doubles as the fresh-identifier supply — `gen`, `genForField`, and `genForLiteralCommitment` (`:256-293`) each return `TVarID(version)` and increment. (Skolem type parameters are* not *drawn from it — they have their own counter, U2 below.) Consequence: within one linear execution segment, distinct* values *states have distinct versions (Lemma 9);*
- origins — *diagnostic provenance for variables (`setOrigin`); no semantic content;*

- *four caches, all part of the saved state:*
  - seenSub $\subseteq$ TType $\times$ TType — *a* success memo *for* ***TType::isSub****: a pair is inserted on acceptance (****TTypeSubtyping.sk:1840****) and consulted before re-checking (****:1837****). Not version-keyed;*
  - seenInter, seenUnion : (TType $\times$ TType) $\rightharpoonup$ TType — *result memos for* ***TType::inter*** *(****TTypeInter.sk:107-123****; pairs containing a bare variable are excluded from the memo) and* ***TType::union*** *(****TTypeUnion.sk:29-36****);*
  - seenExpansion : ExpansionKey $\rightharpoonup$ TType — *memo of one-level named-type expansion, keyed by the dedicated* ***ExpansionKey*** *value class (unqualified name text plus arguments,* ***TType.sk:3055-3057****; read/write* ***TType.sk:697-700****); the cached body is* symbolic *in the type arguments (arguments occur as free variables, re-resolved at every use), which is why reuse under growth is sound (comment* ***TTypeSubst.sk:17-30****). The key's identity premise (name text $\Rightarrow$ definition) and cross-mention exactness are registry rows owned by the expansion paper, not re-argued here.*

save() *snapshots all seven fields into a frozen* ***SavedSubst****;* ***restore()*** *reinstates all seven (****TTypeSubst.sk:36-48****). Because every field is an immutable persistent map/set,* ***save*** *is* $O(1)$ *and lossless.*

impl: ***TTypeSubst.sk / SubstState, SavedSubst***

2.2 Unsaved state

Five pieces of mutable or threaded state live *outside* `SubstState`; each needs its own safety argument.

(U1) `freezeCandidates`**.** Declared on `Subst` itself, not on `SubstState` (`TTypeSubst.sk:76-89`), with an explicit comment that save/restore never touch it; it is push/popped around the constraint-admissibility check in `growTVars` (`TTypeSubtyping.sk:1666-1678`). Verified by inspection: those two lines are the *only* writes to the flag in `src/`, the push/pop brackets a straight-line region with no early return (the `break` at `:1676` exits only the inner constraint loop), and nested checks restore the outer value via `wasFreezing`. So excluding it from `SavedSubst` is safe: a restore can never observe a torn flag, because every save/restore pair either lies entirely inside or entirely outside a freeze window.

(U2) `nextSkolemId`**.** The counter supplying skolem type-parameter identifiers (`TTypeSubst.sk:90-93`, drawn by `genSkolemTParam`, `:262-281`). Deliberately outside `SubstState` *because* restore rewinds the version: reusing a skolem id on a restored timeline would merge two distinct hidden type arguments, so the supply must survive rollback. Exclusion is safe for the same reason it exists — the counter only increases, so ids are never reused across any sequence of saves and restores. Clone-side minting is impossible by construction: `clone()` poisons the field to -1 (`TTypeSubst.sk:160-165`) and `genSkolemTParam` fails hard on a poisoned counter (`:267-272`). Exactly one mint site exists (`TEnv.sk:801`, inside `makeSkolemTArgsOf`), reached from `instanceof` narrowing on the main substitution (`TypeCheckRefine.sk:367`). Skolems are type *parameters* (name-based), not `TVar`s: minting one writes nothing into ρ.

(U3) `ppNames`. A diagnostic-rendering accumulator (`TTypeSubst.sk:94-101`), *shared* by `clone()` on purpose — pretty-printing runs through clones and would otherwise lose the emissions it exists to collect. Safe because it is a record of output only: nothing reads it back into typing, and save/restore never touch it.

(U4) `TEnv`-threaded state. `TEnv` is a *frozen* Skip class; every apparent mutation `!tenv.f = ...` is a functional update of the *local* binding, propagated only into callees that receive the updated `tenv` — so each field below is *path-scoped*: an entry is visible exactly in the dynamic extent of the recording activation and vanishes when it returns. The judgement-relevant fields:

- `tenv.seenSub`, keyed by (version, u_1, u_2) (`TEnv.sk:116`; recorded at `TTypeSubtyping.sk:1877`) — the coinduction hypothesis set; path-scoping does most of the work in the #446 argument (§7.3).
- `tenv.seenInter`, keyed by the bare pair (`TEnv.sk:111`; `TTypeInter.sk:154-157`) — safe because its only effect is to return an *unexpanded* intersection on a cycle, a representation choice, not a judgement.
- `tenv.subtypingHeadDepth` (`TEnv.sk:129`): a per-path count of how often each constructor *pair* has been revisited on the current structural-subtyping descent; past `kMaxSubtypingHeadDepth` = 12 (`TTypeSubtyping.sk:19`) the recursion is *refused* as excessively deep (TS2589, `TTypeSubtyping.sk:1859-1875`). Threaded exactly like `tenv.seenSub`, hence path-scoped; it needs no monotone-reads or rollback argument because a hit can only *reject* — refusal is conservative for soundness, so the field affects completeness only.
- `tenv.expandInter` (`TEnv.sk:131-135`): a mode flag selecting full versus deferred (`unexpandedInter`) field intersection. Its path-scoping and the containment coverage of the deferred form are the OPEN registry row TENV-EXPANDINTER.

The remaining `TEnv` fields are resolution-side: the cycle sets `seenInstantiate`, `instantiatingTParams`, `typeofValues`, `forbiddenInsts` (`TEnv.sk:82,93,102,106` — each self-scoping by the same frozen-threading discipline, and each able only to stop a resolution, never to accept one) and the mode/diagnostic fields `bindings`, `inTypeDefValidation`, `instantiateAt`, `verbose` (`TEnv.sk:65,143,148,155`). They govern which *types* the environment produces — the expansion paper's scope — not the substitution discipline; they are listed so this inventory covers every non-`SubstState` field on both classes.

(U5) `results` and `context`. The diagnostics/callgraph accumulator `results` is shared by `clone()` (`TTypeSubst.sk:149-168`) and never rolled back (e.g. `recordSubtypingEdge` on `TFun::isSub` success, `TTypeSubtyping.sk:1954-1961`). It carries no typing state, so it is outside ρ; §6 flags the one place this is worth a remark.

2.3 Constraint attachment is a value operation

A natural worry about the main paper's Axiom **L-MONO** would be that adding a constraint to a *shared* variable (`TVar::setConstraint`) shrinks that variable's envelope, so an arbitrary type mentioning it could lose members. The premise of that worry is false — establishing this is the single most consequential fact in this paper, and the axiom's remark in the main paper records it.

Lemma 2 (setConstraint is Functional)**.** *`TVar::setConstraint(ty)` does not modify any state observable by another type term. It returns a* new *`TVar` value carrying the enlarged constraint set; every pre-existing occurrence of the same variable identifier retains its own constraint set.*

Proof (by inspection of the pinned source). `TVar` is a frozen (non-`mutable`) class and `setConstraint` a non-`mutable` method (`TType.sk:2988-2992` and `:3001-3004`):

```
fun setConstraint(ty: TType): TVar {
!this.constraints = this.constraints.set(ty); this }
```

In Skip, field assignment through `!this.f = e` in a frozen-class method is a *functional record update of the local binding* `this`; the method must return the updated copy for the change to be visible at all (the sibling `mapConstraints`, `:3006-3009`, uses the same idiom). Two corroborations: (i) `constraints` participates in `TVar::xxhash` (`TType.sk:3010-3019`) and in `Orderable` equality — a shared-mutation semantics would break hashing of interned type terms; (ii) both call sites use the return value (`TType::var(tvar.setConstraint(ty2))`, `TTypeInter.sk:145-151`; `!tvar = tvar.setConstraint(...)`, `TEnv.sk:830-832`) — a genuine mutation would make that redundant. □

Corollary 3 (Occurrence-Relative Envelopes)**.** *The "constraint intersection" step of `Subst::get` (`TTypeSubst.sk:237-239`) intersects the binding with the constraints* of the occurrence being read. *A variable identifier therefore has no global constraint set; constraints are part of the type term. Refinement, intersection, or any other operation attaching a constraint produces a new term and cannot shrink the denotation of any type already installed in an environment. The state ρ of Definition 1 is the* complete *shared mutable typing state.*

2.4 The extended transition table

Definition 4 (Extended Transitions)**.** *The operations that evolve ρ, extending the main paper's Definition 6 (its rows are unchanged and not repeated):*

op	*effect on ρ*	*anchor*
grow, get(cov) *writes*	*per main paper;* version++ *before each write*	`TTypeSubst.sk:170-`
gen/genFor∗	*fresh id =* version*;* version++*; no binding*	`TTypeSubst.sk:256-`
genSkolemTParam	*skolem id from `nextSkolemId` (U2); no ρ effect*	`TTypeSubst.sk:262-`
cache insert	*add to one of the four memo maps; no version bump*	*§1*
setOrigin	*origins only*	`TTypeSubst.sk:311`
save/restore	*snapshot / reinstate all seven fields*	`TTypeSubst.sk:36-`
clone	*fork: new `Subst`, same seven fields, shared `results`/`ppNames`, poisoned skolem counter*	`TTypeSubst.sk:149-`
freeze push/pop	*toggle `freezeCandidates` (unsaved)*	`TTypeSubtyping.sk:`

Verified write-site completeness: the only assignments to `values` in `src/` are in `grow`, `get`, and `restore`; the only version writes are those plus `gen*`; external `grow` callers (`TArrayLitTyping.sk:147,352`; `TVariadicTyping.sk:125`; `TypeCheckUtils.sk:317,328,339`) go through the same transitions.

Note that cache inserts do *not* bump the version. This is sound for the three result memos by their own stability arguments (sealing for `seenInter`, `TTypeInter.sk:90-96`; symbolic arguments for `seenExpansion`) but it makes the version-keying of `tenv.seenSub` track *binding* state only — exactly what the coinduction hypothesis depends on (§7.3).

3 The Order-Theoretic Core

All four ledger rows reduce to statements about one partial order on states and two lemmas about it.

Definition 5 (Envelope; State Order)**.** *For a state ρ and identifier α, the* envelope *is*

$$\mathcal{E}_\rho(\alpha) = \begin{cases} \emptyset & \alpha \notin \mathrm{dom}(\rho.\mathsf{values}) \\ [\![t]\!]_\rho & \rho.\mathsf{values}(\alpha) \in \{\mathsf{SVOpen}(t), \mathsf{SVClosed}(t)\}. \end{cases}$$

Write $\rho \sqsubseteq \rho'$ ("ρ' extends ρ") iff $\mathcal{E}_\rho(\alpha) \subseteq \mathcal{E}_{\rho'}(\alpha)$ for every α. The value $\emptyset$ for unbound variables is the canonical TVAR-SEM definition owned by the main paper; it is the modeling counterpart of get(contra) *returning $\bot$ without writing (`TTypeSubst.sk:230-231`).*

Definition 6 (Modeling Conditions)**.** ***(M1)*** Semantic monotonicity. *For every type term u, $[\![u]\!]_\rho$ is monotone in the envelopes of the identifiers occurring in u: $\rho \sqsubseteq \rho'$ implies $[\![u]\!]_\rho \subseteq [\![u]\!]_{\rho'}$. This holds for the main paper's semantics by structural induction, because every state-sensitive position is read positively: variable occurrences denote $\mathcal{E}_\rho(\alpha) \cap \bigcap [\![\mathrm{cstr}]\!]$ (Corollary 3); field membership constrains only the* get *type (main paper, object semantics, which makes set types opaque); call/new signatures are opaque to $[\![\cdot]\!]$ except for the callability mode; and union/intersection/clause structure is positive. The classic contravariance counterexample is deflected by signature opacity, exactly as the main paper notes for* ***L-MONO****.*

(M2) Union monotonicity. ***L-UNION*** *at the point of use: $[\![t_2]\!]_\rho \subseteq [\![t_2 \vee t]\!]_\rho$.*

Lemma 7 (All Transitions Are Monotone)**.** *Under M2, every operation of Definition 4 except* `restore` *takes ρ to some ρ' with $\rho \sqsubseteq \rho'$.*

Proof. Case analysis on the main paper's transition table plus Definition 4. Fresh bind (unbound $\to$ $\mathsf{SVOpen}(t)$ or $\mathsf{SVClosed}(t)$): envelope grows from $\emptyset$. Fresh covariant get: unbound $\to$ $\mathsf{SVClosed}(\top)$: grows to $[\![\top]\!]$. Open union ($\mathsf{SVOpen}(t_2) \to \mathsf{SVOpen}(t_2 \vee t)$, or $\mathsf{SVClosed}(t_2 \vee t)$ for non-candidates): grows by M2. Seal ($\mathsf{SVOpen}(t) \to \mathsf{SVClosed}(t)$): envelope unchanged. No-op rows, `gen` (the new identifier is unbound before and after — its id was never allocated before, so no term mentions it), cache inserts, `setOrigin`, freeze toggles: envelopes unchanged. No transition removes or replaces bound material: there is no row that shrinks a bound, and the main paper's never-widen lemma additionally freezes SVClosed rows. Envelopes of *other* variables are untouched by each write (values is updated pointwise), though their envelopes may grow transitively via M1 when their bound mentions α — still monotone. □

Lemma 8 (Restore Safety)**.** *Call a computation segment* well-bracketed *if every `restore(s)` it executes uses a snapshot `s` taken by the same activation (verified for all fourteen sites in §7). Let a well-bracketed segment start at ρ_0 and execute monotone transitions and restores. Then at every moment, $\rho_0 \sqsubseteq \rho_{\mathrm{now}}$.*

Proof. Induction over the event sequence. Monotone steps preserve the invariant by transitivity. A restore jumps to a snapshot taken earlier *within the segment* (well-bracketing), and the invariant held when the snapshot was taken, so it holds after the jump. □

Lemma 9 (Record-Point Coherence)**.** *Let a moment m_0 (the* record point*) occur at the* entry *of an activation A, with $\rho.\mathsf{version} = v$ at m_0, and consider any later moment m inside A's dynamic extent, under well-bracketing. Then (i) $\rho.\mathsf{version} \geq v$ at m, and (ii) if $\rho.\mathsf{version} = v$ at m, then $\rho.\mathsf{values} = \rho.\mathsf{values}$ at m_0.*

Proof. (i) The version increases at writes and `gen`, and decreases only at restores. Under well-bracketing, every restore executed inside A's extent uses a snapshot saved inside A's extent, hence saved *after* m_0 (m_0 is A's entry), hence with version $\geq v$. So no reachable restore target has version $< v$. (ii) By induction on the events between m_0 and m. All version-v snapshots taken in the extent are taken at moments where, by the induction hypothesis, `values` equals its m_0 state (a change since m_0 would have bumped the version above v, and by (i) it cannot have come back down except by restoring a version-v snapshot, itself coherent by induction). A moment with version v is either reached with no intervening `values` write since the last coherent version-v moment, or by restoring a coherent version-v snapshot. □

Finding 10 (Failed proof attempt, kept for honesty)**.** *The naive claim "`version` uniquely identifies `values` across a whole checking run" is* false*: restore rewinds the counter, so divergent exploration branches can reuse a version number with different bindings (save at v, grow to $v{+}1$, restore, grow differently to $v{+}1$). Any argument that treats the version as a global state hash — including a reading of the #446 fix as one — is unsound. The actual safety of `tenv.seenSub` comes from the* combination *of version-keying with path-scoping, Lemma 19 below.*

4 L-MONO: Refinement-Time State Monotonicity

L-MONO (the main paper's axiom of that name) demands: for every evolution $\rho \to \rho'$ performed while refining a condition, and every $u \in \text{range}(\Gamma)$, $v \in [\![u]\!]_\rho \implies v \in [\![u]\!]_{\rho'}$.

4.1 Enumeration of refinement-triggered evolutions

By a walk of every arm of `evalCond` (`TypeCheckRefine.sk:476-663`) and its handlers, the evolutions of the *main* substitution that condition refinement can trigger are exactly:

(E1) *Covariant expansion in the refinement primitive.* `setLocalFilterMap` runs `mapSTypes(..., Some(Covariant()), f)` on the main substitution (`TypeCheckRefine.sk:721,736`); expansion applies `subst.get(tenv, tvar, Covariant())` to every spine variable (`expandToVectors`, `TType.sk:680-747`): fresh variables seal to $\top$, open variables seal at their bound. Used by *every* rule that factors through the primitive.

(E2) *Union/intersection during residue and clause rebuilding.* `mapSTypes` reassembles clauses with `Clauses::unionN` (`TType.sk:457`); the `typeof == "object"` default builds `STypes::union` on the main substitution (`TypeCheckRefine.sk:275`). Union itself performs no `get`/`grow` (verified: no such call in `TTypeUnion.sk`) but recursively calls `TType::inter` on field set-types (`TTypeUnion.sk:102,124`), reaching E3.

(E3) *Main-substitution intersections.* `evalIsArray`'s true branch runs `TObject::inter` (`TypeCheckRefine.sk:344`) and `evalInstanceof`'s non-matching branches run `TFields::inter` against the skolem view (`TypeCheckRefine.sk:407`, readonly fields preserved via `keepReadonlyFrom`), both on the main substitution. Inside `TType::inter_`: covariant expansion of both operands (`TTypeInter.sk:158-159`, seals as in E1); constraint attachment on bare-variable operands (`TTypeInter.sk:145-151`, a value operation by Lemma 2); speculative concrete `isSub` probes (`:160-190`); and `TFun::inter`'s two `TFun::isSub` attempts on the main substitution (`TTypeInter.sk:643, 647`) — on success their `growTVars` growth *commits*, on failure it is restored.

(E4) *Fresh identifier generation.* `evalInstanceof` instantiates the class view with fresh *skolem* arguments on the main substitution (`makeSkolemTArgsOf`, `TEnv.sk:780-806`) — a `nextSkolemId` draw with no ρ effect (U2) — and mints ordinary inference arguments only on the throwaway *probe clone* (`TypeCheckRefine.sk:373-379`; `makeTArgsOf`, `TEnv.sk:822-836`: `gen` + occurrence-level `setConstraint`), analyzed as C4 in §6; object-literal typing may create synthetic index variables (`toTFields`, `TypeCheckExpr.sk:112-120`; `createSyntheticIndex`, `TType.sk:1969-1975`).

(E5) *Cache inserts* in all of the above.

Everything else refinement does runs on a *clone* (`TypeCheckRefine.sk:89,99,177,197,373,398,694`) and does not evolve the main ρ at all; those sites are §6's subject.

4.2 Discharge

Theorem 11 (L-MONO)**.** *Under M1 and M2 (Definition 6), every evolution E1–E5 satisfies the* ***L-MONO*** *conclusion — in fact for* all *type terms u, not only* range(Γ)*.*

Proof. E1–E5 are compositions of the transitions of Definition 4 and internal save/restore pairs (inside the speculative `isSub` calls of E3). All fourteen save/restore sites are well-bracketed (§7), so Lemma 8 applies to the whole composition: $\rho \sqsubseteq \rho'$. M1 turns this into $[\![u]\!]_\rho \subseteq [\![u]\!]_{\rho'}$. Two of the main paper's worries dissolve: (a) the `setConstraint` threat is void by Lemma 2 — no shared envelope shrinks, so the restriction to range(Γ) is not needed; (b) the E3 commits from a *successful* speculative `TFun::isSub` are ordinary monotone grows, so "exploration that succeeds and keeps its growth" is safe, not merely "exploration that fails and rolls back." □

Finding 12 (The sharpest sufficient condition is not needed)**.** *The task of bounding* which *variables refinement may touch ("refinement only constrains variables introduced by the very expression being refined") was pursued and abandoned: it is both false — `evalIsArray`/`evalInstanceof` intersect against $\Gamma(x)$'s own object branches, sealing variables shared with the whole definition's checking — and unnecessary, since sealing and growing are monotone for* every *sharer. The variable-sharing pattern is irrelevant once Lemma 2 removes the only non-monotone candidate operation.*

Residual dependence. Theorem 11 is a genuine discharge of **L-MONO** *given the main paper's semantics*: M1 is a property of that semantics (provable by structural induction, sketched in Definition 6; it deserves a short formal proof when the semantics section is next revised), and M2 is **L-UNION**, already a ledger row. The ledger edit in §8 states this precisely.

5 Monotone Reads Under Growth

The obligation: a judgement validated under ρ must still hold after inference evolves $\rho \to \rho'$. For SVClosed bindings this is never-widen(iii); the open question is SVOpen bindings, read without sealing by get(contra).

5.1 The invariant

Definition 13 (Read Disciplines)***(MR)*** Monotone-read invariant. *Every contravariant (non-sealing) read is a* supertype-side *read: its result is used only as the right-hand side of a validation $[\![\text{lhs}]\!] \subseteq [\![\cdot]\!]$, so that later growth of the variable only enlarges the right-hand side.*

(BR) Bake-requires-seal. *Every covariant read whose result is embedded in a persistent artifact (a memoized result, a type installed in an environment) seals the variable, so the baked snapshot equals the envelope at all later states (never-widen(iii)).*

Theorem 14 (Validated Judgements Are Stable)**.** *Assume M1, MR, BR, and* ***L-SUB****. If* `Clauses::isSub` *accepted* $[\![u_1]\!]_\rho \subseteq [\![u_2]\!]_\rho$ *and* $\rho \to \rho'$ *by the transitions of Definition 4 (no restore crossing the acceptance), then* $[\![u_1]\!]_{\rho'} \subseteq [\![u_2]\!]_{\rho'}$*.*

Proof. Decompose the occurrences of variables in u_1, u_2 by how validation read them. Left-side occurrences were read covariantly and are sealed on the success path (BR); never-widen(iii) freezes their envelopes, so $[\![u_1]\!]$ is unchanged... more precisely non-increasing only through frozen envelopes, hence unchanged. Right-side occurrences were read contravariantly (MR); their envelopes only grow (Lemma 7), and by M1 $[\![u_2]\!]$ only grows. The accepted inclusion is therefore preserved. □

5.2 Site audit

Expansion applies its variance uniformly to every *spine* variable, including through `TSInst` and `TInter` unfolding (`expandToVectors`, `TType.sk:680-747`); it does not enter fields or signatures, so no variance flip arises inside a single expansion — flips happen only where the recursion of `isSub` swaps sides.

BR holds at covariant sites *by construction*: get(cov) seals on every path (`TTypeSubst.sk:207-228`) except the freeze-candidate arm (`:221-222`) — the one designed exception, which is this section's finding. Representative covariant sites (the population has grown with the checker and is not usefully enumerable per line; the load-bearing ones are): `isSub_` lhs u_1 (`TTypeSubtyping.sk:1896` — memo `seenSub` baked on success), `TFun::isSub_` rest on the *super* side (`:2145`), both `TTypeInter` operands (`TTypeInter.sk:158-159` — the seal justifies `seenInter`), the `TypeCheckUtils` truthiness/void reads (`:19,42`), the `ArrayLitTyping` commitment-site checks (`TArrayLitTyping.sk:339,365`), and `expandTVarsForPp` (`TType.sk:750`) — which runs on a *clone* (`TTypePp.sk:1202,1215`) precisely so pretty-printing never seals.

The *non-sealing* reads are the exhaustive obligation, and they are: the contravariant sides of validation — `isSub_` rhs u_2 (`TTypeSubtyping.sk:1897`) and `TFun::isSub_` rest on the *sub* side (`:2081`), each used only as the right-hand side of a check (MR); the `resolveSpreadToTuple` shape probes (`TVariadicTyping.sk:51-78`), bracketed by save (`:51`) and restored on every `None` path (`:64,70,78`); the freeze-candidate arm of get(cov) (Finding 15); and the two live-substitution `peekRaw` sites — the spread probe just cited (`TVariadicTyping.sk:67`, inside the restored bracket) and the reflexive-`TSInst` fast path (`TTypeSubtyping.sk:1817`), which is outside this family and analyzed in deep dive D. (`peekRaw` at `TTypeSubtyping.sk:1389,1424` and `TypeCheckExpr.sk:1753` runs on clones.)

No site violates MR: all non-sealing reads either end on the right of a validation or are restored. BR holds at all sites on the success path, with the freeze-arm exception, which is the section's finding.

5.3 Residual hazard: the freeze-window unsealed read

get(cov) on an open *candidate* under `freezeCandidates` returns the bound *without sealing* (`TTypeSubst.sk:221-222`) — a covariant read that violates BR by design, so that constraint admissibility checks are queries. The check runs full `TType::isSub(boundTy, cstr)` (`TTypeSubtyping.sk:1670`), whose *successes* insert pairs into the persistent, version-*unkeyed* `subst.seenSub` memo.

Finding 15 (Stale `seenSub` memo via freeze-window reads)**.** *Candidate scenario: a variable α (an ordinary candidate,* $\mathsf{SVOpen}(t)$*) occurs in* both *sides of a check — e.g. $u_1 = \mathrm{Box}\langle\alpha\rangle$ and u_2 mentions a constrained variable whose constraint check validates* $\mathrm{isSub}(u_1, \mathrm{cstr})$ *under freeze, reading α at t* unsealed *and memoizing* $(u_1, \mathrm{cstr}) \in \mathsf{seenSub}$*. After the freeze pops,* ***growUncovered*** *may contravariantly grow the same α (it is no longer protected) to $t \vee s$. The memo now asserts an inclusion validated with the smaller $[\![u_1]\!]$; a later syntactically-identical query hits it (**TTypeSubtyping.sk:1837**) without re-validation. The eventual covariant expansion of u_1 seals α at $t \vee s$ —* after *the memo was recorded. The discipline itself cannot close this gap: the memo survives on the enclosing check's success path, and nothing re-keys it by version. Mitigations narrow the window without closing it (memos recorded under a speculative subtree are erased by its restore, since* $\mathsf{seenSub}$ *is in **SubstState**; the scenario needs the same open candidate reachable from both sides plus an exact term-level requery). The gap is real: an end-to-end witness exists — an order-dependent unsound acceptance in which* adding *an earlier call to a generic function (validating its constraint under freeze against the still-open candidate, and memoizing the success) flips a correct contravariance rejection of the post-widening call into an acceptance, so a **number**-typed binding holds **"s"** at runtime. MR-1 (§8) is therefore* VIOLATED*. Fix: key **subst.seenSub** by version, or suppress memo insertion while **freezeCandidates** is set; the same fix should cover **seenInter**, which the witness also exercises. Issue draft: **issues/freeze-window-stale-memo-caches.md**; a pinned test is recommended.*

Theorem 14 discharges the obligation *conditional on* MR-1 and on **L-SUB**, and the audit above discharges its MR/BR premises for every other site of the audited `Subst::get`/grow family. One read site sits outside that family: the reflexive-`TSInst` fast path in `TType::isSub` reads an open candidate via `peekRaw` — an exact-value read with no seal, no memo, and no `get`-mediated variance — analyzed and witnessed in deep dive D (issue draft `issues/reflexive-tsinst-fastpath-unsealed-read.md`). The obligation therefore stands VIOLATED at the freeze-window memo and at that read, and PROVED everywhere else.

6 L-CLONE: Clone–Compute–Install

6.1 The pattern and what is actually lost

`evalVarEquality` computes $u = \mathsf{inter}(\Gamma(x), \tau_e)$ under `subst.clone()` and installs u into the scope, discarding the clone (`TypeCheckRefine.sk:87-94`); likewise `removeConstant` (`:96-106`), `evalFieldEqualityType` (`:176-201`), and `evalFieldTruthiness` (`:694`). `clone()` forks all seven `SubstState` fields (immutable maps, so the fork is a true snapshot), shares `results`, `context`, and the output-only `ppNames`, and poisons the skolem counter (U2) (`TTypeSubst.sk:149-168`).

Write ρ_f for the fork state (= the main state at the clone), ρ_c for the clone state when `inter` returns, and ρ'' for any later main state. What the clone loses at discard, classified by what `inter` can do:

(C1) *Constraint attachments* (`TTypeInter.sk:145-151`). By Lemma 2 these never touched the clone's state at all: the constraint lives in the returned occurrence inside u, which is installed. *Nothing is lost.* A natural conjecture ("if constraints live in the returned type, clone-discard may be benign for constraints but not for value bindings") is confirmed on its first half.

(C2) *Seals* from covariant expansion (`TTypeInter.sk:158-159`): the expansion replaces the spine occurrence by the read value in u — fresh variables bake $\top$, open variables bake their fork-state bound t — but the seal itself dies with the clone; on the main substitution the variable stays open/unbound and may later grow or seal differently.

(C3) *Grows and commitments* from speculative subtyping inside `inter`: `TFun::inter` runs `TFun::isSub` both ways on the clone (`TTypeInter.sk:643, 647`), committing growth on success; `ArrayLitTyping` may commit a literal's form variable on the clone. All lost at discard, while u may mention the affected variables.

(C4) *Fresh generations* on the clone (the `instanceof` preservation probe mints inference arguments on its probe clone, `TypeCheckRefine.sk:373-379`; expansion of a cold `TSInst` can reach `tenv.instantiate` and, through object-literal branch building, `gen`-sites such as synthetic index fields, `TType.sk:1969-1975`). The clone's version counter advances privately; the main counter does not.

6.2 Over-approximation: the positive result

Theorem 16 (Clone Stability)**.** *Assume M1, M2, and the fork-state form of* ***L-INTER****:*

(I$_f$) $[\![t_1]\!]_{\rho_f} \cap [\![t_2]\!]_{\rho_f} \subseteq [\![u]\!]_{\rho''}$ for every $\rho'' \sqsupseteq \rho_f$, *where* $u = \mathsf{inter}$ *computed by a well-bracketed run from* ρ_f.

Then ***L-CLONE*** *holds: the installed u over-approximates the fork-state intersection under the main substitution at install time and at all later states. Moreover I_f follows from the ordinary* ***L-INTER*** *plus M1 for every value whose membership does not depend on clone-side state that u bakes or mentions; the two ways such dependence can arise are exactly C2/C3 occurrences and C4 identifiers, analyzed below.*

Proof sketch. The clone evolves monotonically from ρ_f (Lemma 7 + Lemma 8): $\rho_f \sqsubseteq \rho_c$. The main substitution also evolves monotonically: $\rho_f \sqsubseteq \rho''$. Ordinary **L-INTER** gives containment in $[\![u]\!]_{\rho_c}$; the gap is that ρ_c and ρ'' are *incomparable* (divergent branches of ρ_f), so M1 cannot bridge them directly — this was the first failed proof attempt, and the reason **L-CLONE** cannot be reduced to **L-INTER** + M1 alone. The bridge is per-occurrence: (C2) material *baked* into u is the clone-state envelope of the variable at read time, which by monotonicity of the clone run is $\supseteq$ its fork-state envelope; baked material is state-independent thereafter, so reading u under ρ'' yields at least what reading it with fork-state envelopes would — and the fork-state intersection is contained in that by **L-INTER** *evaluated at* ρ_f. (C1) occurrences carry their constraints with them and denote $\mathcal{E}_{\rho''}(\alpha) \cap \bigcap [\![\text{cstr}]\!]$, which by $\rho_f \sqsubseteq \rho''$ contains the fork-state value — the intersection semantics is exact here. (C3) growth affects u only through signature positions (`TFun::inter` returns one of the two function terms, whose parameter/return types may mention clone-grown variables), and signatures are opaque to $[\![\cdot]\!]$ — the main paper's signature-opacity deflection applies verbatim. Field-get positions reach `inter` recursively and fall under C1/C2. What the sketch does *not* cover is C4, next. □

6.3 Clone-escaped fresh identifiers: an unreachable hazard

Finding 17 (Clone-minted identifiers never escape into installed types)**.** *Fresh identifiers are drawn from the* version counter *(`TTypeSubst.sk:256-293`), and a clone forks the counter: if a clone generates a variable (C4), its id equals the id the* main *substitution assigns to its own next generation, and a clone-generated occurrence escaping into an installed type in a semantically visible position (a field get type — synthetic index fields qualify, `TType.sk:1969-1975`) would, once the main substitution binds the colliding id, silently read an unrelated variable's envelope. Nothing in Theorem 16's bridge would cover that: the occurrence's denotation would not*

be monotone from fork state. The hazard is settled by an escape audit of the `gen*` *call sites. Exactly one runs under a clone: the* `instanceof` *preservation probe mints inference arguments on* `probeSubst` *(*`TypeCheckRefine.sk:373-379` *via* `makeTArgsOf`*,* `gen` *at* `TEnv.sk:829`*). Its products are confined by construction:* `probeFields` *is consumed only as the right-hand side of per-branch* `TFields::isSub` *probes, each on a fresh re-clone (*`TypeCheckRefine.sk:395-402`*), and the branches actually* installed *are either the scrutinee's own (*`fields`*) or intersections computed on the main substitution from main-substitution material (*`TypeCheckRefine.sk:407-415`*) — no probe-clone identifier occurs in an installed type. Every other* `gen*` *site (*`TypeCheckFun.sk:318`*;* `TArrayLitTyping.sk:195`*;* `TypeCheckUtils.sk:315,326,337,344`*;* `TypeCheckExpr.sk:4485`*; the two* `createSyntheticIndex` *callers in* `toTFields`*,* `TypeCheckExpr.sk:112-120`*) runs on the expression-typing substitution, never under a refinement clone — re-verified by enumerating all* `.gen(`*/*`.genForField(`*/*`.genForLiteralCommitment(` *sites at the pin. So no clone-side fresh identifier reaches a type installed against the main substitution, and CL-1 (§8) is* discharged*. The discharge rests on a code invariant that should be guarded: assert (and CI-check) that clone-minted variables do not escape the probe; for the skolem class the analogous guard already* is *a runtime invariant —* `clone()` *poisons* `nextSkolemId` *and* `genSkolemTParam` *fails hard on it (*`TTypeSubst.sk:160-165, 267-272`*).*

Shared `results`**.** Clone-side speculative `TFun::isSub` successes record subtyping edges into the shared accumulator even when the exploration is discarded (`TTypeSubtyping.sk:1954-1961`). This does not touch typing state; it can only over-connect the callgraph consumed by later phases. Out of scope for type soundness, recorded here so the boundary is explicit.

7 No Commitment on Failure: the Rollback Audit

The governing contract is stated at the top of the subtyping module (`TTypeSubtyping.sk:4-7`): *every* `isSub` *function must guarantee that when it returns* `STError`*, the subst is unchanged.* This section verifies it, pairs every save with its restores, and documents the one place the letter of the contract is violated by design.

7.1 Exhaustive save/restore pairing

All `save`/`restore` sites in the pinned revision (`grep` over `src/`), each verified well-bracketed (restore uses the same activation's snapshot; no early exit escapes the pair):

#	site (save; restore)	restore condition
1	`TFields::isSub` (`TTypeSubtyping.sk:943; 945`)	on `STError`
2	`STypes::isSub` (`:1307; 1334`)	on failure
3	`Clauses::isSub` (`:1347; 1367`)	on overall failure (clause with no matching partner)
4	`TObject::isSub` outer (`:2183; 2195`)	on overall failure
5	`TObject::isSub` per branch pair (`:2187; 2192`)	after each failed pairing, before trying the next
6	`TType::isSub_` (`:1890; 1916`)	on validation failure (growth persists on success, intentionally)
7	`growTVars` per-variable (`:1651; 1682`)	on constraint-check failure
8	`isCovered` probes (`:1589; 1600`)	after *every* probe, hit or miss
9	`TFun::isSub` (`:1949; 1952`)	on `STError`
10	`ArrayLit` tuple commit (`TArrayLitTyping.sk:402; 416`)	on element-check failure (case 2c)
11	`resolveSpreadToTuple` (`TVariadicTyping.sk:51; 64,70,78`)	on every `None` path
12	`inferRestTypeVar` (`TVariadicTyping.sk:124; 130`)	on constraint failure (error also reported)
13	`satisfies` (`TypeCheckExpr.sk:3218; 3227`)	*unconditionally* — `satisfies` must not commit inference
14	clone-as-rollback: `TEnv.sk:378`, `TypeCheckRefine.sk` (7 sites), `TypeCheckExpr.sk` (7 sites), pretty-printer (`TTypePp.sk:1202,1215`)	discard instead of restore

Three structural corollaries. (i) Well-bracketing holds at all sites, so Lemmas 8 and 9 apply globally. (ii) `SavedSubst` captures *all* mutable typing state (Definition 1); the exclusions are exactly U1–U5 of §2 — `freezeCandidates` (safe by its push/pop bracketing), `nextSkolemId` (safe *because* it is never rewound), `ppNames` (output-only), the `TEnv`-threaded fields (path-scoped; `tenv.seenSub` by §7.3), and `results`/`context` (no typing state). In particular a rollback also erases `seenSub`/`seenInter`/`seenUnion`/`seenExpansion` entries recorded in the failed subtree, so no stale *success* memo survives a rollback. (iii) Because no environment binding is written between any save and its restore (refinement installs happen strictly after the operations return), a restore reinstates a state under which every installed type was already satisfied — rollback needs no analogue of **L-MONO**.

7.2 The intentional exception: array-literal commitments

`ArrayLitTyping::subtyping` runs at the top of `TType::isSub_` *before* the save at `:1890` (`TTypeSubtyping.sk:1884-1888`). Its cases 1 and 2b commit the literal's form variable and then return the result of a subsequent check *even when that check fails* (`TArrayLitTyping.sk:339, 365-392`): the commitment persists on an `STError` return, violating the module contract's letter. This is deliberate ("irreversible", `:140`); case 2c by contrast restores the commitment on failure (row 10).

Lemma 18 (The carve-out is sound for accepted programs)**.** *A commitment persisting on a failed check cannot cause unsound acceptance, provided every caller that* discards *an* ***`STError`*** *(uses* ***`isSub`***

as a probe) either wraps the call in save/restore or cannot reach an uncommitted array literal.

Proof, with caller audit. On a non-discarded failure the program is rejected — the committed type may be wrong for the runtime value (case 2b commits to `Array<expected>` even when the elements fail against `expected`), but no judgement about an accepted program uses it. The risk is a discarded failure. Probe callers: `isCovered` (row 8) restores around every probe; `growTVars` constraint checks (row 7) restore on failure; the speculative probes inside `inter_` (`TTypeInter.sk:160-190`) and `SType::overlaps` (`:393`) compare a boxed-primitive instance against a single `SType` — neither side is a bare uncommitted `TArrayLit`, which is the only trigger shape (`getSingleArrayLitUncommitted` requires a sole `TArrayLit` inst, `TArrayLitTyping.sk:68`); `TFun::inter`'s two attempts run under `TFun::isSub`'s own save/restore (row 9), which was taken *before* the inner `isSub_` ran, hence before the commitment — a failed first attempt rolls the commitment back. The constraint probe at `TEnv.sk:378` runs on a clone (row 14). □

This audit is complete for the pinned revision but fragile — any new `isSub`-as-probe without a save/restore wrapper re-opens it. Sub-obligation RB-1 in §8 records the maintenance condition.

7.3 `tenv.seenSub`: the #446 argument, made precise

`tenv.seenSub` is the coinduction set of `isSub_`: on entry, if $(\mathsf{version}, u_1, u_2)$ is present the check succeeds by coinductive hypothesis; otherwise the triple is recorded and threaded into the recursive calls (`TTypeSubtyping.sk:1852, 1877`). It is not part of `SubstState`, and no rollback ever touches it. Its safety rests on two independent mechanisms:

Lemma 19 (Version-Keyed Hypotheses Are Inert or Exact)**.** *A lookup that hits an entry (v, u_1, u_2) observes* values *identical to the recorder's state at record time. Consequently a hypothesis is either consulted under exactly the binding state it was made in, or (after any growth, seal, or fresh generation, each of which bumps the version) it is* inert *— present but unmatchable.*

Proof. Two mechanisms. *Path-scoping* (U2, §2): `TEnv` is frozen, so the entry is visible only in the dynamic extent of the recording activation — entries never leak across sibling explorations or survive returns, which already rules out the cross-branch collision of Finding 10. *Record-point coherence*: the record happens at the activation's entry, so Lemma 9 applies to every lookup in the extent: version $= v$ implies `values` equal to record time. (The lemma's premise — restores in the extent use snapshots from the extent — is row-by-row verified in §7.) □

This is the precise content of the #446 regression class and its guard: without the version key, a rolled-back exploration's hypothesis could be consulted under different bindings *within the same extent* (probe loops, rows 7–8, restore and re-grow inside one `isSub_` activation); the version key makes exactly those entries inert. The prerequisite "version is bumped on every `values` write" is verified at all three write sites (`TTypeSubst.sk:200,214,224`); note `gen*` bumps too, which additionally protects hypotheses against fresh-variable aliasing inside an extent.

By contrast, `subst.seenSub` — the success memo, *not* version-keyed — is protected against rollback staleness by living inside `SubstState` (corollary (ii) above), and against growth staleness by Theorem 14; it is the memo that inherits the MR-1 hazard (Finding 15).

8 Status Table and Ledger Edits

Named sub-obligations introduced by this paper:

MR-1 No `subst.seenSub` memo recorded while `freezeCandidates` is set may outlive subsequent growth of an open candidate readable from its left-hand side (Finding 15). Status: VIOLATED, witnessed end-to-end (issue draft `issues/freeze-window-stale-memo-caches.md`). Fix options: version-key the memo, or skip memo insertion under freeze; either must also cover `seenInter`. Pinned test recommended.

CL-1 No fresh variable generated on a clone may occur in a type installed against the main substitution (Finding 17). Status: discharged — the property holds by the escape audit: the single clone-side `gen*` site (the `instanceof` preservation probe) confines its products to probe-side subtype checks, and every other `gen*` site runs on the expression-typing substitution; guard with the recommended CI assertion (clone-minted variables must not escape the probe — the skolem analogue is already a runtime invariant).

RB-1 Maintenance condition: every `isSub` caller that discards `STError` must wrap the call in save/restore or be unreachable by uncommitted array literals (Lemma 18); a CI grep for new probe sites is the cheap enforcement.

M1 Semantic monotonicity of $[\![\cdot]\!]$ in variable envelopes (Definition 6) — a property of the main paper's semantics, provable by structural induction; should be added there as a lemma rather than assumed here.

What this paper establishes, row by row:

Row	**Established here**
L-MONO	Theorem 11, conditional on M1 and L-UNION; the `setConstraint` threat is void (Lemma 2) and the range(Γ) restriction is droppable
Obl. Monotone Reads	general statement via Theorem 14 (given M1 + the MR/BR audit, §5.2); the residue MR-1 is witnessed (Finding 15), and the reflexive-`TSInst peekRaw` read sits outside the audited family (deep dive D)
L-CLONE	Theorem 16 given I_f (fork-state L-INTER; registry row I-F); the id-collision hazard CL-1 discharged by the non-escape audit of the single clone-side `gen*` site (Finding 17), CI-assertion guard recommended
Obl. No Commitment on Failure	14-row save/restore audit (§7; `SavedSubst` completeness; `freezeCandidates` and `tenv.seenSub` exclusions justified, Lemma 19); array-literal commitments persist on failure by design, sound by Lemma 18 under RB-1

Registry. The authoritative, current statuses are maintained in the registry (`docs/ledger.yaml`), rendered as the main paper's Assumptions and Proof Obligations tables; this paper is cited there as evidence. The registry rows it supports:

Row	Statement	Status
L-MONO	Legal inference-state changes during refinement preserve membership in types computed earlier.	PROVED
L-CLONE	A type computed by `inter` under a cloned substitution over-approximates the intersection under the main one.	PROVED
OBL-MONOTONE	Monotone Reads Under Growth: judgements validated under ρ survive later legal transitions.	VIOLATED
OBL-RESTORE	No Commitment on Failure: failing and exploratory paths restore all relevant state.	PROVED
MR-1	No `subst.seenSub` memo recorded under `freezeCandidates` may outlive growth of an open candidate readable from its left side.	VIOLATED
CL-1	No fresh variable generated on a clone occurs in a type installed against the main substitution.	PROVED
M1	Semantic monotonicity of $[\![\cdot]\!]$ in variable envelopes (a property of the get-only, signature-opaque semantics).	ASSUMED
I-F	Fork-state form of L-INTER: the containments hold from a forked substitution state (consumed by L-CLONE's proof).	ASSUMED
RB-1	Maintenance condition: `isSub` callers discarding errors are save/restore-wrapped or unreachable by uncommitted array literals.	OPEN
SETCONSTRAINTFUN	`TVar::setConstraint` is a functional update on a frozen value: constraints are occurrence-level, not shared state.	PROVED
E-TRANSPORT	Monotone-reads transport: an inclusion validated under ρ survives later legal transitions on the same timeline, except for the cache behavior recorded by MR-1.	PROVED
TENV-EXPANDINTER	The `tenv.expandInter` mode flag is path-scoped, and both intersection forms it selects (expanded and deferred `unexpandedInter`) satisfy the L-INTER containment.	OPEN

Imports (review contract). The named statements this paper assumes from elsewhere. A reviewer treats each as an axiom, checking citations against the registry's canonical statement; no other paper's internals are in scope:

Row	Canonical statement	Owner	Status
L-UNION	`union` over-approximates set union (all levels).	B	PROVED
L-INTER	`inter` over-approximates set intersection (the containment clause).	B	PROVED
L-SUB	`Clauses::isSub` acceptance implies semantic inclusion under the current state.	D	OPEN
TVAR-SEM	A type-variable occurrence denotes its substitution envelope intersected with its occurrence constraints; an unbound envelope is empty.	main	PROVED

Summary. Three of the four rows close (L-MONO; No Commitment on Failure, with a documented carve-out; L-CLONE modulo I_f, its id-collision hazard verified unreachable and guarded by a CI assertion). The fourth — Monotone Reads — is PROVED at every audited site except its one sharp residue, MR-1, which is VIOLATED with an end-to-end witness (issue draft `issues/freeze-window-stale-memo-caches.md`); the fix (version-keying or freeze-window write suppression) must cover both `seenSub` and `seenInter`. A further violated read site, the reflexive-`TSInst` fast path's `peekRaw`, lies outside this paper's audited family and is analyzed in deep dive D (issue draft `issues/reflexive-tsinst-fastpath-unsealed-read.md`).

A Implementation Anchors

Pinned to `bafdcf7c`.

Model	Implementation
ρ (seven fields)	`SubstState` (TTypeSubst.sk:13–49)
snapshot / reinstate	`SavedSubst`, `save`, `restore` (TTypeSubst.sk:36–74)
transitions	`Subst::grow/get` (TTypeSubst.sk:170–241)
fresh ids from version	`gen/genForField/genForLiteralCommitment` (TTypeSubst.sk:256–293)
skolem ids (unsaved counter)	`nextSkolemId`, `genSkolemTParam` (TTypeSubst.sk:90–93, 262–281)
pp-name record (unsaved)	`ppNames` (TTypeSubst.sk:94–101)
freeze (unsaved)	`Subst.freezeCandidates` (TTypeSubst.sk:76–89); push/pop TTypeSubtyping.sk:1666–1678
clone fork	`Subst::clone` (TTypeSubst.sk:149–168)
occurrence constraints	`TVar`, `setConstraint` (TType.sk:2988–3019)
success memo	`subst.seenSub` (TTypeSubtyping.sk:1837,1840)
coinduction hypotheses	`tenv.seenSub` (TEnv.sk:116; TTypeSubtyping.sk:1852,1877); `tenv.seenInter` (TEnv.sk:111; TTypeInter.sk:154–157)
head-depth refusal	`tenv.subtypingHeadDepth` (TEnv.sk:129; TTypeSubtyping.sk:19,1859–1875)
result memos	`seenInter/seenUnion/seenExpansion` (TTypeInter.sk:107–123; TTypeUnion.sk:29–36; TType.sk:697–700)
uniform-variance expansion	`expandToVectors` (TType.sk:680–747)
validation spine	`isSub_/growTVars/isCovered` (TTypeSubtyping.sk:1562–1916)
clone-compute-install	`evalVarEquality` etc. (TypeCheckRefine.sk:87–106,176–201,694)
main-subst refinement ops	`setLocalFilterMap` (TypeCheckRefine.sk:721,736); `evalIsArray/evalInstanceof` (TypeCheckRefine.sk:326–420)
form commitments	`ArrayLitTyping` (TArrayLitTyping.sk:140–148,326–432)
external grow callers	TArrayLitTyping.sk:147,352; TVariadicTyping.sk:125; TypeCheckUtils.sk:317,328,339